

CHAITANYA CHAKKA

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EDUCATION

Boston University

Sep 2024 - Exp. Dec 2025

Masters in Artificial Intelligence | GPA: 4.00/4.00

- **Image Video and Computing**
- Natural Language Processing
- Multimodal Machine Learning

Birla Institute of Technology and Science Pilani Hyderabad Campus

Aug 2019 – May 2023

Bachelor of Engineering in Computer Science, Minor in Data Science | CGPA: 3.76/4.00

THESIS AND PUBLICATIONS

Improving Prompt Alignment in Vision Language Models: A Self-Learning Framework for Generative Models

Manuscript in progress - Tianle Chen, **Chaitanya Chakka**, Deepti Ghadiyaram

- Developed a self learning framework for any text-to-image generation models including Stable Diffusion, Janus and Imagen to improve object state generation for passive prompts.
- Engineered synthetic training data generation pipeline with automated evaluation to quantify emptiness/absence of objects.

Social Implications of OCEAN Personality: An Automated BERT-Based Approach, ICAART (3), 2024.

Chakka, Chaitanya, Bathini Sai Akash and Aruna Malapati. [Publication](#)

- Created a prediction pipeline leveraging classic ML models(SVM, LR) and complex models like MLP and BERT with Bi-LSTM head integrating 80 psycholinguistic features with LLM embeddings, achieving high accuracy in OCEAN personality trait prediction.
- Conducted a case study on correlation of demographics with personality traits utilizing the Mann-Whitney U statistical test.

Interpreting LLM Representations: Analyzing mBERT's encoding of Code-Switched Data, Research Square, 13 December 2023

Chaitanya Chakka, Mithun Kumar S R, Aruna Malapati, [Preprint](#)

- Crafted English-Spanish code-switched datasets for token-level probing tasks for better transformer interpretability.
- Constructed a layer-wise probing framework with perceptron classification head to analyze linguistic encoding across mBERT layers, identifying optimal layers for syntactic and semantic tasks in code-switched contexts using MCC and F1 metrics.

EXPERIENCE

Cashfree Payments

June 2023 – June 2024

Software Development Engineer

Bangalore, India

- Spearheaded integration of Kong API Gateway with Golang server wrappers across 30 teams serving 1500 TPS daily. Revamped the system with custom Lua plugins for team-specific use cases and deployed in Kubernetes with Postgre SQL backend.
- Introduced Twilio for Whatsapp and SMS notification channels via gateway routing, improving delivery rates by 20%.
- Reviewed and approved API gateway changes, building integration tests for deployment pipelines to ensure 100% uptime.

Smible and Co.

Oct 2021 – Feb 2022

Technical Co-founder

www.tinyurl.com/smibbleOfficial

- Created and launched the MVP of a voice-first socio-educational networking app on the Google Play store. Formulated a scoring system to enable transparent credibility. Built on Flutter, Google Firebase backend, and Agora RTC for voice calling.
- Reached 150+ users with 40-50% daily engagement fostering collaboration in domains including Philosophy and Technology.

PROJECTS

Optimizing LLM Question Generation for Enhanced Conversational AI | LLM, Question Answering, [Github](#)

Oct 2024 – Dec 2024

- Constructed a modular framework with context aware answer validation resulting in a 5% more efficient question answering.
- Developed multi-stage pipeline leveraging BERT and GPT for context-aware follow-up question generation, improving interpretability scores by 3%.

Quantitative Analysis of Instruction Specificity in Vision-Language Navigation | Computer Vision, NLP, Multimodal ML

Feb 2025 – Present

- Engineered a novel computational metric to quantify instruction vagueness in vision-language navigation models.
- Implemented statistical analysis revealing high specificity variance across ALFRED, SPOC and Room2Room datasets.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, Golang, Lua, Bash scripting

Libraries: Transformers, Diffusers, PyTorch, Tensorflow, OpenAI API, OpenCV, Pillow, scikit-learn, Matplotlib, LangChain

Frameworks and Tools: Echo, Kong, AWS, Git, HPC, Remote Sensing

ADDITIONAL EXPERIENCE

Graduate Teaching Assistant | Dr. Andrew Wood | Foundations of Data Science - Boston University

Jan 2025 – Present

- Teaching crucial math-intensive foundational topics in Data Science for a class of 90 students at Boston University.
- Delivered personalized feedback to 90 students, resulting in a 30% improvement in average assignment scores and a noticeable increase in class participation during lectures on EM and PCA.